Why Domain Matters: shelf, truck and money, on one map

Who saw what

The record says

`The records agree.`

What actually happened

Eight of twelve joins compare a record with its own copy.

Three failures, one month, every test green

The shelf

5 kg atta in the wrong bay

10–13 June. 71 bags not sold.

`Availability on the books: 97%`

The truck

1 kg packs on a 5 kg pallet line

22–27 June. 72 bags not sold.

`22 June: on time, in full`

The money

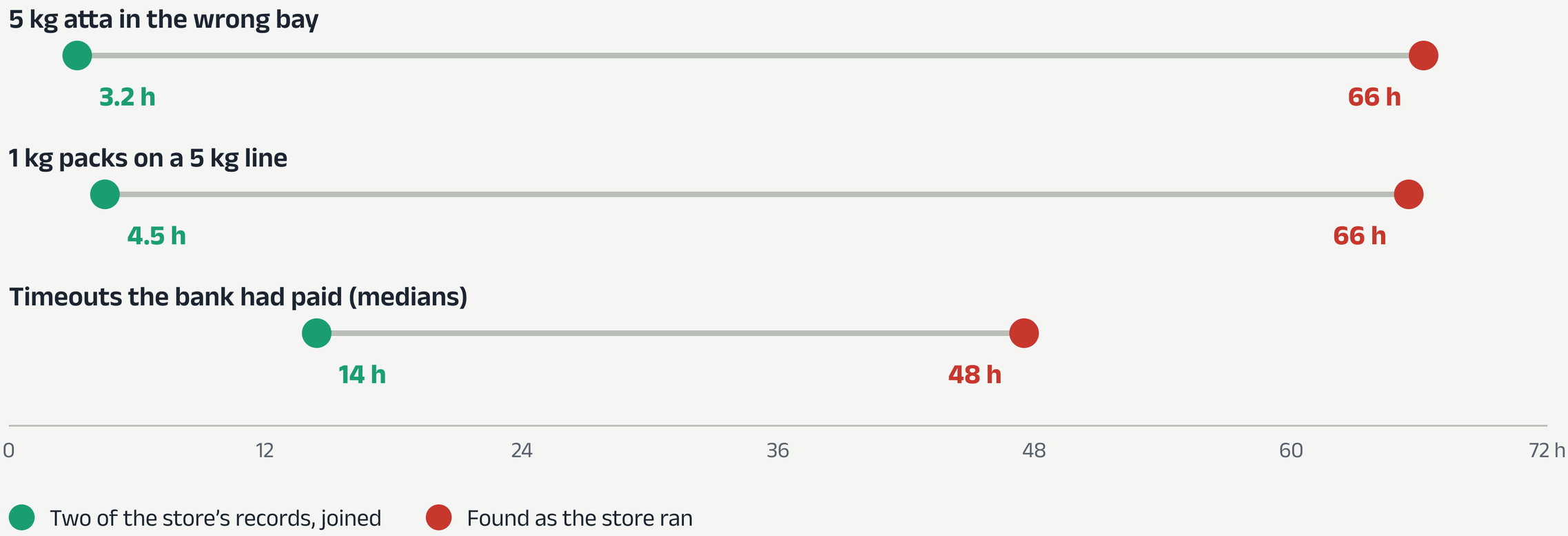
Timeouts the bank had already paid

4–24 June. ₹13,366.06 taken twice from 24 customers.

`Every bill paid exactly once`

Each time, the dashboards were green, the data tests passed, and an AI assistant given the schema said nothing was wrong.

Hours by joining, days by waiting



Hours after the shelf emptied, or after the extra debit. 3 of the customers charged twice never complained.

Join the records two at a time

Eight always agree: copies

70 of 70	shipping notice lines match the pick confirmations
70 of 70	goods receipts match the notice
297 of 297	item-days of book stock equal the sum of receipts, sales, write-offs and counts
18,384 of 18,384	approved UPI payments match the provider’s report

Four sometimes disagree

4 of 25	stock counts agree with the books
86 of 88	shelf audits agree with the books
9 of 10	receipts survive the next count
16 of 48	timeouts agree with the provider’s report

12 joins between Store 014’s records in June. The eight that agree hold 32,754 rows between them, and never disagree.

A test on a copy can only pass.

The notice is written from the picks, the receipt by scanning a label against the notice, the book stock from the receipts and the sales. That is why every pipeline in this series was green.

Each of June’s three failures sits on one of the four joins that disagree. On each, one side looks at the world.

Who saw what

	The truck	The shelf	The money
The KPI	**OTIF**	**On-shelf availability**	**Payment success**
reads	Goods receipts: a copy of the notice (70 of 70)	Book stock: a running sum (297 of 297)	Till payments: the provider’s answer, or a timeout (18,384 of 18,384)
contradicted by	Stock counts: 9 of 10 receipts survive	Counts 4 of 25; audits 86 of 88	Provider’s report: 16 of 48 timeouts agree
which looks at	The shelf and the backroom	The shelf	The customer’s bank account

Nothing looked at what was on a pallet. None of June’s 30 pallets was weighed or case-scanned.

Across the shelf and the truck

Store, Tue 23 Jun, 20:00	**Till lines and book stock** `No 5 kg sale since 16:00; 9.9 expected; 63 on the books`
Store, Mon 22 Jun, 06:12	**Goods receipt GRN-260622, line 1** `64 bags booked by scanning the pallet label against the notice`
Warehouse, Mon 22 Jun	**Shipping notice SHP-260622, line 1** `16 cases of 5 kg atta on pallet …0542`
Warehouse, Mon 22 Jun, 03:00	**Pick confirmation** `16 cases from slot A-07-03, by a slot label scan`
Warehouse, Tue 23 Jun, 14:20	**Correction, slot A-07-04** `1 kg atta: −16 cases (−160 packs), “short pick: slot count variance”`

Line 1 of SHP-260622 held 1 kg atta: 160 packs. Right.

Named at 20:00 on Tuesday: 38 hours before the warehouse’s pair, 61 before the store’s count.

Neither domain’s rules get there alone

The retail rules

The scripted agent asked for a shelf check at 22:00 on Tuesday. On Wednesday morning it found the shelf and the backroom empty, corrected the book and reordered: 22 bags lost instead of 72.

It never learned why. The 160 extra packs of 1 kg, the warehouse’s slots and the delivery still counted in full were outside its rules.

The logistics rules

Pair the warehouse’s corrections: equal and opposite, neighbouring slots, within a week.

The second correction comes on Thursday at 10:00, 38 hours after the map could have named the line.

The agent is scripted, so the rerun can be repeated: same customers, same trucks, seed 42.

What the ledger showed

Failure	What it cost	What the ledger showed
5 kg atta in the wrong bay	71 bags not sold	Nothing. No record holds a sale that did not happen.
1 kg packs on a 5 kg pallet line	72 bags not sold	Four stock corrections in two buildings, netting to ₹80 at cost (₹44,648 gross).
Timeouts the bank had already paid	₹13,366.06 taken twice	UPI credits like any other. ₹10,520.00 went back after complaints; ₹2,846.06 was still in suspense when June closed.

Bags at the shelf price customers would have paid. Corrections at cost, as the store books them.

Give the agent the map

3 → 21

of 21 questions right, schema only → with domain layers, across the three posts

214 → 105

bags of 5 kg atta lost in an average June, a replenishment agent reading the KPIs → reading them with domain rules

16% → 54%

GPT-4 on an insurer's business questions: from the SQL tables → from a knowledge graph of the business (data.world, 2023)

The series' assistant and agent are scripted so every run can be repeated. Sequeda, Allemang and Jacob, arXiv:2311.07509.

A domain layer is the map, as data

1. Every record, the event that writes it, and what it looks at in the world
2. Every join, and how often it agrees
3. Every KPI, the record it reads, and the observation that can contradict it
4. The rules for reading a silence or a timeout as a question

`kvinayakpai.github.io/domain-series/data/india-grocery/store-014-map.json`

Store 014's map, with the SQL for every join, next to the records it describes.

Five questions to ask your team

1. For each KPI an agent acts on, which record does it read, and is that record a copy?
2. Which of our joins have never disagreed, and do we know why?
3. What in our business does no record look at?
4. When two departments' records disagree, who joins them, and how soon?
5. Can our agents read the map: what each record sees, and what can contradict it?

For data and AI leaders.

Every record was right about what it saw. Join the ones that look at the world.

The series, the KPIs x-rayed, the records and the map:

kvinayakpai.github.io/domain-series

> **One store, one month**
>
> The shelf, the truck and the money. Eighteen KPIs, each measured twice.